

Yichen YAN

Curriculum Vitae

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Education

- 2024–present **BSc, Computing & Software Systems**, *The University of Melbourne*, Melbourne, Australia.
Machine Learning, Computer Systems, Software Modelling and Design, Probability for Statistics, Database Systems, Discrete Maths & Operations Research
- 2026 **Exchange, Business Intelligence & Data Analytics**, *Zhejiang University*, Zhejiang, China.

Research Interests

Artificial Intelligence; Machine Learning; Computer Vision; UAV-based Perception; Object Tracking; Transfer Learning; Robotics Perception..

Research Experience

2023 **Robustness and Real-time Improvement for Vehicle Tracking by UAV based on CamShift Algorithm with Hue, Saturation, Value Components.**

- Prepared an unpublished manuscript titled *Robustness and Real-time Improvement for Vehicle Tracking by UAV based on CamShift Algorithm with Hue, Saturation, Value Components*.
- Investigated vehicle tracking from UAV-mounted cameras using CamShift, MeanShift, HSV colour components, search windows, and two-dimensional histograms.
- Explored methods to improve tracking robustness and real-time performance under complex urban backgrounds.
- Gained early exposure to computer vision, UAV-based perception, object tracking, literature review, and academic writing.

Advisor : **Dr. Peng Lu**, *Assistant Professor, Department of Mechanical & Engineering*, The University of Hong Kong ([My Research](#))

Academic Projects

2026 **Image Classification with Transfer Learning and Classical Machine Learning Models**, *The University of Melbourne – COMP30027 Machine Learning Project*.

- Developed a complete machine learning pipeline for coarse-grained animal classification and fine-grained bird species classification.
- Compared hand-crafted visual descriptors, including colour histograms, HOG descriptors, and low-level image statistics, with ImageNet-pretrained CNN embeddings.
- Used ResNet and EfficientNet variants as fixed feature extractors, converting images into high-dimensional learned representations without fine-tuning on the project data.
- Evaluated Logistic Regression, Linear SVM, RBF SVM, KNN, Gaussian Naive Bayes, Random Forest, and Extra Trees across different feature combinations.
- Applied stratified train-validation splits and 5-fold stratified cross-validation for model selection and hyperparameter tuning.
- Achieved 97.3% validation accuracy and 97.3% macro-F1 on coarse-grained animal classification using HOG, additional features, and EfficientNet-V2-L embeddings with Logistic Regression.
- Achieved 97.6% validation accuracy and 97.5% macro-F1 on fine-grained bird species classification using HOG, additional features, and EfficientNet-V2-L embeddings with Random Forest and Extra Trees.
- Conducted fine-grained error analysis using confusion matrices, identifying visually similar bird species pairs as the main source of misclassification.

- 2026 **Financial Asset Trend Analysis and Watchlist Monitoring Platform**, *Independent Vibe-Coding Development Project*.
- Developed an investment research platform for Chinese stocks and mutual funds using AI-assisted coding tools.
 - Implemented indicators including moving averages, RSI, MACD, and Bollinger Bands to support trend analysis.
 - Designed rolling backtesting modules to evaluate future 5-, 20-, and 60-trading-day win rates, average returns, excess returns, and maximum drawdowns.
 - Built a market scanning and ranking engine with PostgreSQL caching to generate explainable trend-strength rankings.

Professional Experience

- Jan 2026 – **Internet Sector Analyst Intern**, *Huatai Securities*, Shanghai, China.
- Apr 2026
- Benchmarked AI agents and video-generation models by designing structured prompts and evaluation dimensions across reasoning, instruction-following, multimodal capability, content quality, and commercial applicability.
 - Developed scoring frameworks to translate qualitative AI model outputs into structured analyst-ready comparisons.
 - Synthesised AI product trends, model capabilities, and competitive positioning into research materials for internet-sector analysis.
- Jul 2025 – **Financial Advisory Intern**, *Meritco Services*, Shanghai, China.
- Jan 2026
- Supported deal sourcing and pre-investment due diligence across early-stage AI, embodied-intelligence, robotics, and consumer technology startups.
 - Prepared business plans and financing materials by analysing technical roadmaps, market size, product positioning, revenue models, competitive landscape, and investment highlights.
 - Assisted investor communication workflows for financing projects, including meeting preparation, meeting minutes, feedback tracking, and follow-up materials.
- Jun 2024 – **Equity Research Analyst Intern**, *Sealand Securities*, Shanghai, China.
- Oct 2024
- Supported equity research in electronics, semiconductor, and AI-related technology sectors by tracking industry data, company announcements, and supply-chain developments.
 - Prepared structured research materials covering business models, revenue structure, customer base, industry cycle, growth drivers, and potential risks.

Academic Achievements & Recognitions

- 2022 **Global Gold Award**, British Physics Olympiad
- 2022 **Regional Top Individual and Group Award**, National Economics Challenge.

Computer skills

Programming Languages	Python, Java, C, SQL/MySQL, R, Matlab
Machine Learning	Image classification, transfer learning, feature extraction, model evaluation, Logistic Regression, SVM, KNN, Naive Bayes, Random Forest, Extra Trees, scikit-learn
Software and Tools	Git, Jupyter Notebook, LaTeX, basic Linux command line, PowerPoint, Wind, Choice
AI Tools	ChatGPT, Claude, Codex, Qwen, prompt design, AI model benchmarking, research synthesis

Leadership and Activities

- Jun 2024 – **Secretary, Organisation Department**, *Chinese Students Association, The University of Melbourne*, Melbourne, Australia.
- Jul 2025
- Coordinated inter-university and campus events with 50–300+ attendees.
 - Managed meeting logistics, internal communications, and cross-team workflows for the student committee.
 - Recognised as an Outstanding Student Leader by the association.